

Elite Quant Brainteaser Assessment - Revised Full Paper

Questions Q1-Q60 | Brainteasers & Logic Tests | Questions Only

Q1. A constrained Tower of Hanoi has pegs A-B-C-D in a line; disks may move only between adjacent pegs. Disks 1 through 4 start stacked on A and must end stacked on D. Standard size legality applies. What is the minimum number of moves?

- A. 30
- B. 34
- C. 38
- D. 42
- E. Impossible

Q2. A pile has 29 stones. Players alternate removing 1, 3, or 4 stones. A player may not remove the same number as the immediately previous move. The player who removes the last stone wins. With optimal play, what should the first player remove?

- A. 1
- B. 3
- C. 1 or 3
- D. 4
- E. No winning first move

Q3. Six traders A, B, C, D, E, F sit around a circular table. Each either always tells the truth or always lies. Each says: "My two neighbors are of different types." Exactly two of the six are liars. How many distinct type assignments are possible up to rotation but not reflection?

- A. 0
- B. 1
- C. 2
- D. 3
- E. 6

Q4. A state is a permutation of 1,2,3,4,5. A legal move reverses either the first three entries or the last four entries. Starting from 12345, what is the minimum number of moves to reach 15342?

- A. 3
- B. 5
- C. 6
- D. 7
- E. Impossible

Q5. You have seven sealed boxes numbered 1 through 7. Exactly three contain red balls and four contain blue balls. You may ask yes/no questions of the form: "Among boxes in subset S, is the number of red balls odd?" An adversary answers consistently but wants to maximize the number of questions you need before identifying all red boxes. What is the minimum worst-case number of questions needed?

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Q6. A modified Tower of Hanoi has three pegs A, B, C and four disks. Initially all disks are on A; the goal is all disks on C. Standard size legality applies, but the pegs have capacities: A holds at most 4 disks, B holds at most 2 disks, and C holds at most 4 disks. Additionally, disk 1 cannot be moved twice consecutively. What is the minimum number of moves?

- A. 15
- B. 17
- C. 19
- D. 21
- E. Impossible

Q7. A game begins with a marker at 0 on a line. Alice and Bob alternate moves; Alice starts. On a turn, a player adds either 2, 5, or 7 to the marker. The player who first makes the total strictly exceed 40 loses. However, exactly once in the game, Bob may instead subtract 1, but only if the current total is odd. With optimal play, what is Alice's first move?

- A. Add 2
- B. Add 5
- C. Add 7
- D. Alice wins with any first move
- E. Alice loses with any first move

Q8. Four switches control four lamps, but the wiring is unknown: each switch controls exactly one lamp and each lamp exactly one switch. You may set switches, enter the lamp room, observe which lamps are on/off, and feel whether any off lamp is warm. You may enter the room only twice. Before the first entry, all lamps are cold and off. Assuming lamps become detectably warm after being on for long enough and remain warm for one entry interval after being turned off, what is the maximum number of possible wirings that can be distinguished with certainty?

- A. 16
- B. 20
- C. 22
- D. 24
- E. Impossible to distinguish all wirings in two entries

Q9. Five coins lie in a row, each showing H or T, initially H T H T H. In one move you may choose any consecutive block of length 2 or 3 and flip every coin in that block. Your goal is T T T T T. However, after every move, the leftmost coin showing H, if any, must be strictly farther right than it was immediately before the move. What is the minimum number of moves?

- A. 2
- B. 3
- C. 4
- D. 5
- E. Impossible

Q10. There are eight cards numbered 1 through 8 face down. A hidden permutation is chosen. You may ask comparison questions of the form: "Is the card at position i larger than the card at position j ?" You win once you can identify the exact position of card 8. An adversary answers consistently with some permutation and wants to delay you. What is the minimum worst-case number of comparisons required?

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Q11. A row of 10 lamps is initially off. One move toggles a consecutive block of length 3 or length 5. What is the minimum number of moves needed to reach the pattern 1010010110?

- A. 3
- B. 4
- C. 5
- D. 6
- E. Impossible

Q12. Four piles are arranged in a cycle with counts (12,0,0,0). A move chooses adjacent piles with unequal counts and transfers one token from the larger pile to the smaller pile. What is the minimum number of moves needed to reach (3,3,3,3)?

- A. 10
- B. 12
- C. 14
- D. 16
- E. Impossible

Q13. A game: players alternately remove 1-3 stones; removing last loses. Starting with 17 stones, what is the optimal first move?

- A. Remove 1
- B. Remove 2
- C. Remove 3
- D. No winning move
- E. Remove 4

Q14. A token starts at 0 modulo 11. Exactly 7 moves are made. Each move adds 1, 4, or 6 modulo 11, and the same increment may not be used on consecutive moves. How many valid move sequences end back at 0?

- A. 2
- B. 4
- C. 6
- D. 8
- E. 10

Q15. In a grid, movement is allowed only if the parity of the coordinates changes on each unit step. From (0,0) to (5,5), what is the minimal number of steps?

- A. 10
- B. 9
- C. Impossible
- D. 8
- E. 12

Q16. Two players alternately pick numbers 1-9 without repetition; the first player whose chosen numbers contain a subset summing to 15 wins. What is the optimal first move?

- A. 1
- B. 3
- C. 5
- D. 7

E. 9

Q17. A 3-agent circular passing game starts with value 1. Each pass doubles the value modulo 7. After exactly 5 passes, what is the value?

A. 2

B. 3

C. 4

D. 5

E. 6

Q18. Choose four distinct primes from {2,3,5,7,11,13,17,19}. How many 4-prime selections sum to 37?

A. 1

B. 2

C. 3

D. 4

E. 5

Q19. A state machine evolves a binary string by simultaneously flipping the neighbors of every 1. Starting from 10101, what is the string after 2 steps?

A. 01010

B. 11111

C. 00000

D. 10101

E. 01110

Q20. Twelve lamps are arranged in a circle, initially all off. One move toggles four cyclically consecutive lamps. What is the minimum number of moves needed to reach the cyclic pattern 111110001000?

A. 1

B. 2

C. 3

D. 4

E. Impossible

Q21. A standard 3-peg Tower of Hanoi has disks 1 through 4 initially on A and target C. Disk 1 is forbidden from moving on move numbers divisible by 3. What is the minimum number of moves?

A. 18

B. 22

C. 26

D. 30

E. Impossible

Q22. A token starts at state 0 on a 7-cycle. On each turn a player may move it +1, +2, or +4 modulo 7, but may not choose the same increment as the previous turn. The player who first moves the token to state 0 after the initial position wins. With optimal play, starting player should open with:

A. +1

B. +2

C. +4

D. Any move wins

E. No opening move can force a win

Q23. Five traders each wear a red or blue badge. Each sees all badges except their own and simultaneously writes one of {R, B, Pass}. They win as a team if at least one non-pass guess is correct and no non-pass guess is wrong. Before seeing badges, they agree on a deterministic strategy. What is the maximum guaranteed number of badge assignments on which they win?

- A. 24 of 32
- B. 25 of 32
- C. 26 of 32
- D. 27 of 32
- E. 28 of 32

Q24. Five boxes in a line contain (0,6,0,6,0) tokens. A legal move chooses an interior box and either moves one token from it to each neighbor, or reverses that operation if both neighbors have at least one token. What is the minimum number of moves to reach (4,0,4,0,4)?

- A. 6
- B. 8
- C. 10
- D. 12
- E. Impossible

Q25. In a constrained Hanoi system with pegs L, M, R, disks may move only along edges L-M and M-R, never directly L-R. Additionally, disk 1 cannot be moved twice consecutively. Starting with disks 1-4 on L, target all disks on R. What is the minimum number of moves?

- A. 26
- B. 30
- C. 32
- D. 34
- E. Impossible

Q26. Two players alternately claim unused numbers from 1 to 12. After each claim, if a player's claimed set contains three distinct numbers with one equal to the sum of the other two, that player immediately loses. If both play optimally, which first claim avoids a forced loss?

- A. 1
- B. 2
- C. 3
- D. 4
- E. No first claim avoids a forced loss

Q27. A black-box machine has four binary switches. In one press you may flip either exactly one switch or exactly three switches. You start at 0000 and must visit every 4-bit state exactly once before returning to 0000. What is the length of the shortest valid cycle, counting the final return move?

- A. 8
- B. 12
- C. 16
- D. 17
- E. Impossible

Q28. Six agents sit around a circle. Initially the coin counts are (6,0,0,0,0,0). A move chooses adjacent agents with unequal coin counts and transfers one coin from the richer to the poorer. Where rotations and reflections are

considered the same, what is the minimum number of moves to reach a state in which all agents have one coin?

- A. 6
- B. 7
- C. 8
- D. 9
- E. Impossible

Q29. A number starts at 1. Each of exactly 5 operations replaces n by either $2n+1$ or $3n-1$. How many operation sequences end at a multiple of 7?

- A. 1
- B. 2
- C. 4
- D. 6
- E. 8

Q30. There are 9 locked doors in a line, initially all closed. A legal action chooses a consecutive block of length 2 or 5 and toggles every door in that block. What is the minimum number of actions needed to open all 9 doors?

- A. 3
- B. 4
- C. 5
- D. 6
- E. Impossible

Q31. Three pegs A, B, C hold disks 1 through 5, all initially stacked on A. Standard size rules apply. In addition, every move of disk 1 must be followed immediately by a move of disk 2, unless disk 1's move is the final move of the entire process. What is the minimum number of legal moves to move the full tower to C?

- A. 37
- B. 41
- C. 45
- D. 49
- E. Impossible

Q32. A row of 8 lamps is initially off. A legal move toggles exactly one adjacent pair, except that the same pair may not be toggled on two consecutive moves. What is the minimum number of moves needed to make exactly lamps 1, 4, 6, and 8 on, if possible?

- A. 4
- B. 5
- C. 6
- D. 7
- E. Impossible

Q33. Two players alternately place one of their own stones on an empty vertex of a regular hexagon. A player loses immediately if their stones contain three vertices with pairwise graph distances 1, 2, and 3 along the hexagon. With optimal play, what is the outcome?

- A. First player wins on move 3
- B. First player wins on move 4
- C. Second player wins on move 3
- D. Second player wins on move 4

E. Draw until the board is full

Q34. Four analysts simultaneously receive hats numbered 0, 1, or 2 modulo 3. Each sees the other three hats, not their own, and must either guess their own number or pass. They win as a team if at least one person guesses correctly and no one guesses incorrectly. They may pre-agree on a deterministic strategy. What is the maximum number of the 81 assignments on which they can guarantee a win?

- A. 54
- B. 60
- C. 63
- D. 72
- E. 80

Q35. A token starts at (0,0). In each step you may add one of (1,2), (2,1), (-1,2), or (2,-1). You must never visit a point whose coordinate sum is divisible by 5, except possibly on the final step. What is the smallest number of steps needed to reach (7,7)?

- A. 4
- B. 5
- C. 6
- D. 7
- E. Impossible

Q36. A pile has 29 stones. Players alternate removing 1, 3, or 4 stones, but a player may not remove the same number as the immediately previous move. The player who takes the last stone wins. With optimal play, how many stones should the first player remove?

- A. 1
- B. 3
- C. 4
- D. Both 1 and 4 work
- E. No winning first move

Q37. You have five boxes in a line containing (0,4,0,4,0) tokens. A legal move chooses an interior box and moves one token from it to each neighbor, or reverses such a move if both neighbors have at least one token. What is the minimum number of moves needed to reach (2,0,4,0,2)?

- A. 4
- B. 5
- C. 6
- D. 7
- E. Impossible

Q38. In a four-peg Hanoi variant, disks 1 through 6 start on peg A and must end on peg D. Standard size rules apply. Disk 6 may move only once. Disk 5 may never be placed on peg B. During the entire process, peg C may never contain more than two disks. What is the minimum number of moves?

- A. 17
- B. 19
- C. 21
- D. 23
- E. Impossible

Q39. A hidden bitstring of length 6 is all zeros. Each query chooses any three positions and flips them. After exactly k queries, the final bitstring must have Hamming weight 2. What is the smallest positive k for which this is possible?

- A. 1
- B. 2
- C. 3
- D. 4
- E. Impossible

Q40. A pile has 63 stones. Players alternate removing 1, 2, 4, or 8 stones. A player may not remove the same number as the immediately previous move. The player who removes the last stone wins. Which first move is winning?

- A. Remove 1
- B. Remove 2
- C. Remove 4
- D. Remove 8
- E. No winning first move

Q41. A four-peg Hanoi system has disks 1 through 6 initially on A and target D. Standard size legality applies. Pegs B and C may each hold at most two disks at any time. What is the minimum number of moves?

- A. 17
- B. 20
- C. 24
- D. 28
- E. Impossible

Q42. A token starts at 0 modulo 13. Exactly 5 moves are made. Each move adds 1, 4, or 6 modulo 13, and the same increment may not be used on consecutive moves. How many valid move sequences end back at 0?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Q43. A 10-bit lock starts at 0000000000. One operation toggles a consecutive block of length 3 or length 5. What is the minimum number of operations needed to reach 1001111001?

- A. 1
- B. 2
- C. 3
- D. 4
- E. Impossible

Q44. A state is a permutation of 1,2,3,4,5. A legal move reverses either the first three entries or the last four entries. Starting from 12345, what is the minimum number of moves to reach 54123?

- A. 4
- B. 5
- C. 6
- D. 7
- E. Impossible

Q45. Two players alternately mark empty cells of a 3 by 4 board. A player loses immediately if after their move their own marks contain the four corners of any axis-aligned rectangle. With optimal play, and assuming both prefer delaying

loss once loss is unavoidable, what happens?

- A. First player can force a win on move 5
- B. First player can force a win on move 6
- C. Second player can force a win on move 5
- D. Second player can force a win on move 6
- E. The board can be filled without either player losing

Q46. There are five piles arranged on a cycle with counts (9,0,0,0,0). A legal move chooses adjacent piles and transfers exactly one token from the larger pile to the smaller pile; if equal, that edge cannot be chosen. What is the minimum number of moves required to reach a configuration where all pile counts differ by at most 1?

- A. 8
- B. 9
- C. 10
- D. 11
- E. 12

Q47. A binary string of length 10 starts as 0000000000. One operation chooses a block of length 3 or 4 and toggles every bit in the block. What is the minimum number of operations needed to reach 1011111101?

- A. 3
- B. 4
- C. 5
- D. 6
- E. Impossible

Q48. A pile has 37 stones. Players alternate removing 1, 2, 4, or 8 stones. A player may not remove the same number as the immediately previous move. The player who removes the last stone wins. Which first move or moves are winning?

- A. Remove 1 only
- B. Remove 2 only
- C. Remove 4 only
- D. Remove 1 or 4
- E. No winning first move

Q49. A robot is at vertex 0 of a cube. Each move traverses an edge, but the robot may not traverse two consecutive edges in the same coordinate direction. It must visit all 8 vertices at least once and return to 0. What is the length of the shortest such closed walk?

- A. 8
- B. 10
- C. 12
- D. 14
- E. Impossible

Q50. A constrained Hanoi line has pegs A-B-C-D in a path; disks may move only between adjacent pegs, never across a missing edge. Disks 1 through 4 start on A and must end on D. Standard size rules apply. Additionally, peg B may never contain exactly two disks. What is the minimum number of legal moves?

- A. 38
- B. 42
- C. 46
- D. 50

E. Impossible

Q51. A four-peg Hanoi system has disks 1 through 7 initially on A and target D. Standard size rules apply. Disk 7 may move only once. Disk 6 may not be placed on B or C until disk 3 has visited D. Disk 1 may not be moved twice in a row. What is the minimum number of legal moves?

A. 25

B. 27

C. 29

D. 31

E. Impossible

Q52. A token starts at 0 on a 13-cycle. On each turn a player adds 1, 2, or 5 modulo 13. The same increment may not be used on consecutive turns, and the player who first returns the token to 0 after the initial state wins. With optimal play, what opening move gives the first player a forced win?

A. Add 1

B. Add 2

C. Add 5

D. Both 1 and 5

E. No opening move forces a win

Q53. A token starts at 0 modulo 19. Exactly 7 moves are made. Each move adds 1, 2, or 5 modulo 19, and the same increment may not be used on consecutive moves. How many valid move sequences end back at 0?

A. 2

B. 4

C. 6

D. 8

E. 10

Q54. A row of 11 lamps is initially off. A legal move toggles exactly one block of length 2, 3, or 6. Blocks may overlap. What is the minimum number of moves needed to turn on exactly lamps 1, 2, 5, 6, 7, 10, and 11?

A. 3

B. 4

C. 5

D. 6

E. Impossible

Q55. Four piles are arranged in a line with counts (12,0,0,0). A move chooses adjacent piles with unequal counts and transfers one token from the larger pile to the smaller pile. What is the minimum number of moves needed to reach (3,3,3,3)?

A. 12

B. 14

C. 16

D. 18

E. Impossible

Q56. Two players alternately color uncolored vertices of a 3-dimensional cube, one vertex per turn, using their own color. A player loses immediately if their colored vertices contain a path of length 3 along cube edges. With optimal play, what is the result?

A. First player loses on their third move

- B. First player loses on their fourth move
- C. Second player loses on their third move
- D. Second player loses on their fourth move
- E. The cube can be fully colored without a loss

Q57. A machine state is a pair (x,y) modulo 8. One operation adds either $(1,3)$, $(3,1)$, or $(5,5)$. Starting at $(0,0)$, you must visit exactly 16 distinct states and return to $(0,0)$ on the next operation. What is the smallest possible number of operations including the final return?

- A. 12
- B. 14
- C. 16
- D. 17
- E. Impossible

Q58. A constrained Hanoi line has pegs A-B-C-D-E in a path; disks move only between adjacent pegs. Disks 1 through 3 start on A and must end on E. Standard size rules apply. Additionally, disk 1 may never move directly from C to D. What is the minimum number of legal moves?

- A. 24
- B. 26
- C. 28
- D. 30
- E. Impossible

Q59. A number starts at 1. Each of exactly 4 operations replaces n by one of $2n+1$, $3n-1$, or $n+5$. How many operation sequences end at a multiple of 8?

- A. 4
- B. 6
- C. 7
- D. 8
- E. 10

Q60. A binary string of length 12 starts at all zeros. One operation chooses a cyclically consecutive block of length 4 and toggles it. What is the smallest positive number of operations needed to reach a string with exactly six 1s, if possible?

- A. 2
- B. 3
- C. 4
- D. 5
- E. Impossible